

Association of midday napping with hypertension in aged over 45 population: a cross-sectional study in China

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Submitted to: JMIR Public Health and Surveillance
on: April 15, 2022

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Abstract

Background: Hypertension is one of the main worldwide public health problems and midday napping is a popular habit. The relationship between the two remains to be explored.

Objective: To explore the association between midday napping and hypertension and test the mediation effect of BMI.

Methods: This study separately selected 12414, 14894, 11244 Chinese respondents aged over 45 from China Health and Retirement Longitudinal Study (CHARLS) 2011, 2015 and 2018 datasets. Binary logistic regression was used to explore the relationship between midday napping and hypertension, and three-step method was used to test the mediation effect of BMI.

Results: Among all the respondents, the prevalence rates of hypertension were 24.6% (2011), 20.9% (2015) and 22.7% (2018). Compared to no-nappers, those napping over 90 minutes had greatest risk suffering from hypertension in 2011 (OR=1.484, 95%CI=1.297~1.698) and 2015 (OR=1.372, 95%CI=1.212~1.553). In 2018, napping 30-90 minutes had greatest risk for hypertension (OR=1.333, 95%CI=1.193~1.49). ORs of different napping duration decreased from 2011 to 2018. Besides, BMI had a partial mediation effect in 2015 and 2018.

Conclusions: Midday napping contributed to hypertension with BMI acting as a mediator. To prevent hypertension, long duration of midday napping need to be avoided and keep BMI at a normal level.

(JMIR Preprints 15/04/2022:38782)

DOI: <https://doi.org/10.2196/preprints.38782>

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Association of midday napping with hypertension in aged over 45 population: a cross-sectional study in China

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Abstract

Background: Hypertension is one of the main public health issues around worldwide and midday napping is a popular habit. The association between the two remains to be explored.

Objective: To explore the association of midday napping with hypertension.

Methods: This study separately selected 11439, 12689, and 9464 Chinese respondents aged over 45 from China Health and Retirement Longitudinal Study (CHARLS) 2011, 2015, and 2018 datasets. Binary logistic regression was used to explore the association of midday napping with hypertension, and the three-step method was used to test the mediation effect of BMI.

Results: Among all the respondents, the prevalence rates of hypertension were 24.64% in 2011, 21.14% in 2015, and 22.10% in 2018. Midday napping was positively correlated with hypertension. In 2011 and 2015, napping 60-90 min had the greatest ORs ($OR_{2011}=1.705$, $OR_{2015}=1.494$). In 2018, the biggest OR fell into the group of napping 30-60 minutes ($OR=1.223$), ORs of different napping duration decreased from 2011 to 2018. Besides, BMI had a partial mediation effect in 2015 and 2018.

Conclusions: Midday napping is a potential risk factor for hypertension with BMI acting as a mediator. To prevent hypertension, prolonged duration of midday napping and actions to maintain a normal BMI level are recommended.

Keywords: hypertension; risk factor; midday napping; BMI; mediation effect

Introduction

Hypertension is one of the main public health issues worldwide and it has been identified as one of the main risks for stroke, heart failure, and cerebrovascular disease [1-3]. As of 2019, 1.28 billion people, or more than 16% of the world's population, suffered from hypertension [4]. It was estimated to contribute to 50% of coronary heart disease cases and two-thirds of the cerebrovascular disease burden [5]. Successive population surveys conducted in China over the last 30 years have revealed an increasing prevalence of hypertension [6, 7]. Now there are 270 million hypertension patients in China and it has become the main culprit for disability-adjusted life years in China, contributing to 24.6% of all-cause mortality [8, 9].

Considering the high prevalence and enormous health toll, a series of actions have been taken in China. In 2009, the New Health Care System Reform was introduced and hypertension management was made a vital public health service free for all patients [10]. It stipulated that primary healthcare facilities must provide residents with free screening, management, and follow-up services [11]. Additionally, the Chinese central government has constructed many national demonstration areas for the comprehensive prevention and control of hypertension to improve the lifestyle, health literacy of the population, and community-based hypertension management [12]. The turning point came when the Primary Health Care, Medicine and Health Promotion Law was implemented in 2020, which is pioneering legislation for health promotion in China. It established the legality and necessity of a population-wide hypertension prevention and control approach [13]. As a result, the long-ignored prevention of hypertension is becoming increasingly crucial, emphasizing that the improvement of modifiable risk factors should be a public priority.

Previous studies have identified some modifiable risk factors related to hypertension, including excessive drinking, smoking, an unhealthy diet and less exercise [14-16]. Some researchers spotted the link between sleep and hypertension, and concluded that sleep duration and quality were strongly associated with the risk of hypertension [17-19]. However, the effect of midday napping, which is another popular sleep activity, has rarely been addressed. Although some studies indicated an independent association between midday napping and the incidence of hypertension [20-22], the results were at odds with each other. Additionally, the association of hypertension with overweight and obesity has been extensively proven and the prevalence of hypertension among the obese population may range from 60% to 77%, increasing with body mass index (BMI) [23, 24]. Prolonged midday napping duration was found to elevate cortisol levels, resulting in abnormal fat distribution [25]. Also, decreased thermogenesis and energy expenditure, and activated sympathetic nervous system (SNS) caused by midday napping may also contribute to obesity [26, 27]. Therefore, BMI seems an appropriate mediator, while exploring the association between midday napping and hypertension, and can help understand the underlying mechanism. Thus, three samples from China Health and Retirement Longitudinal Study (2011, 2015, 2018) were employed in this study to examine the relationship between midday napping and hypertension among middle-aged and older Chinese people and test the mediation effect of BMI. By identifying the potential risk modifiable factors, this study aimed to impose influences on individual lifestyles and public policy to control hypertension.

Materials and Methods

Sample and data collection

The primary data used in this study is from the China Health and Retirement Longitudinal Study (CHARLS), a longitudinal national study conducted in 450 neighborhood/village committees in 150 counties across 28 provinces. A four-stage, stratified, cluster probability sampling design was adopted in the baseline survey and detailed sampling procedures were shown in Wang et al.'s study [28]. The data regarding individuals' demographic and socioeconomic status, health conditions, and related behaviors information were collected among residents over 45 years old in China. Part of the participants were excluded for the following reasons: (1) age less than 45; (2) missing values for BMI or height and weight; (3) missing information about hypertension. The final sample size is 11439 in 2011, 12689 in 2015, and 9464 in 2018.

Variables

Primary dependent variable

The dependent variable was a binary variable, indicating whether a resident suffered from hypertension. Hypertension is defined in accordance with the national guidelines for primary hypertension prevention and management [29, 30]: without anti-hypertensive drugs, systolic blood pressure is over 140 mm Hg and/or diastolic blood pressure is over 90 mm Hg, or currently taking anti-hypertensive drugs, or previously diagnosed as hypertensive by a clinician.

Primary independent variable

Midday napping was set as the independent variable, grouped by napping duration, which was appraised using a self-reported questionnaire that asked, "During the past month, how long did you take a nap after lunch on average?". According to existing literature, five groups were defined [31, 32]: (1) no napping. (2) napping time <30 minutes. (3) napping time = 30-60 minutes. (4) napping time = 60-90 minutes. (5) napping time >90 minutes.

Control variables

Sociodemographic characteristics (gender, age, education, residential status, marital status, household annual income per capita), health-related variables (self-reported health status, activities of daily living (ADL), mental health, personal medical histories, Body Mass Index (BMI), lifestyles (smoking status, drinking status and, night sleep duration) were included in this study [33-36]. The information was collected by using a structured questionnaire. Education was classified into 4 groups [33]: 1. Never attended school. 2. Primary school and lower. 3. Junior middle school and lower. 4. Senior middle school and higher. Residential status were classified as rural and urban [34]. Marital status was classified into three groups [37]: 1: Married/Cohabiting, 2: Divorced/Widowed /Separated, 3: Never married. Age and household annual income per capita were set as continuous variables and household annual income per capita was log transformed [38]. Health status was appraised by self-reported poor health: Very good, Good, Fair, Poor, and Very poor [36]. ADL were measured based on Basic Activities of Daily Living (BADL) and Instrumental Activities of Daily Living (IADL). Participants were categorized as ADL impaired if they reported difficulty or

inability to perform any activity item [39]. Mental health was appraised by the 10-item version of the Center for Epidemiological Studies Depression Scale (CESD-10). Participants were classified into No depressive symptoms (CESD score < 10) and Depressive symptoms (CESD-10 score ≥ 10) [40]. Personal medical histories included diabetes and cardiovascular diseases (CVD), which were self-reported by the participants. CVD were defined as self-reported having chronic heart problems, stroke, or both [41]. BMI was categorized into four groups [42]: Underweight (BMI < 18.5), Normal (18.5 $\leq$ BMI < 25.0), Overweight (25.0 $\leq$ BMI < 30.0), and Obese (BMI ≥ 30). Drinking status were classified into three groups according to the frequency of drinking: Drink more than once a month, Drink but less than once a month, and Non-drinker. Smoking status was a binary variable, indicating whether smoked or not. Night sleep duration was categorized into five groups taken in hours [36]: <6, 6-7, 7-8, 8-9, and >9. The details of the variables are shown in Table 1.

Table 1. Definition/codes of variables

Variable	Codes/definition
Hypertension	0= No ☐ 1= Yes
Gender	1=Male ☐ 2=Female
Age (year)	Real value
Education	1=Never attended school ☐ 2=Primary school and lower ☐ 3=Junior middle school and lower, 4=Senior middle school and higher
Residential status	1=Rural, 2= Urban
Marital status	1=Married / Cohabiting, 2=Divorced / Widowed / Separated, 3=Never married
Household annual income per capita (log, yuan)	Real value
Health status	1=Very good, 2=Good, 3=Fair, 4=Poor, 5=Very poor
ADL impaired	1=Yes, 2=No
Mental health	1=No depressive symptoms, 2=Depressive symptoms
Diabetes	1=Yes, 2=No
CVD	1=Yes, 2=No
BMI	1=Underweight, 2=Normal weight, 3=Overweight, 4=Obese
Drink	1=Drink more than once a month ☐ 2=Drink but less than once a month ☐ 3=Non drinker
Smoke	1=Yes, 2=No
Night sleep duration (hour)	1=<6 ☐ 2=6-7 ☐ 3=7-8 ☐ 4=8-9, 5=>9
Midday napping duration (minute)	1=0 ☐ 2=<30 ☐ 3=30-60 ☐ 4= 60-90, 5=>90

Data analysis

Statistical Product Service Solutions (SPSS) software was used to describe the characteristics of participants, and the disparity in hypertension across different groups was examined by chi-square and independent sample t-test. After adjusting for control variables, binary logistic regression was employed to explore the relationship between midday napping and hypertension. The variables in the regression model were selected by the Enter method. The association between midday napping and hypertension was quantified using odds ratios (ORs) having 95 percent confidence intervals (95% CI), with other variables controlled. To verify whether BMI played a role in midday napping's influence on hypertension, the three-step method proposed by Baron was employed to test the

mediating effect of BMI [43]. The judging criteria for whether there was a mediation effect were taken as: (1) There is a statistically significant relationship between the independent variable (X, coefficient=a) and the mediator (M); (2) There is a significant relationship between the independent variable (X, coefficient=c) and the dependent variable (Y); (3) The coefficient of the mediator (coefficient=b) in the regression model which contained the independent variable, mediator and, the dependent variable is statistically significant [44]. The mediator was defined as a complete mediator if the coefficient of X was not significant in the regression model including X, M and Y. And it became a partial mediator if the coefficient of X was still significant in the regression model, indicating other remained factors in the path from X to Y. The mediation effect value was calculated as $a*b$, and the ratio of the mediating effect with the total effect was calculated as $a*b/c$ [45]. $P<0.05$ (two-tailed) was regarded as statistically significant. The data were described and analyzed by SPSS 24.0.

Results

Sample Characteristics

There was a reasonably steady percentage of participants overall who had hypertension: 24.64% (N=2818) in 2011, 21.14% (N=2683) in 2015, and 22.10% (N=2092) in 2018. Participants who regularly took midday naps ranged in number from 6166 (54.19%) in 2011, 7409 (58.39%) in 2015, to 5727 (60.51%) in 2018. Among all the midday nappers, those who napped between 60 and 90 minutes took the biggest portion, with 23.88% (N=2717) in 2011, 27.43% (N=3480) in 2015, 23.63% (N=1717) in 2018. There were slightly more female participants than male (2011: 5421 vs 6018, 2015: 6265 vs 6424, 2018: 4493 vs 4971), with an average age of 59.53 in 2011, 61.00 in 2015, and 60.58 in 2018. More information could be seen in Table 2.

Table 2. Characteristics about participants

		2011 (N)				2015 (N)				2018 (N)			
		Total	No hypertension	Hypertension	P ^a	Toal	No hypertension	Hypertensio n	P ^a	Total	No hypertension	Hypertension	P ^a
		N=11439	N=8621	N=2818		N=12689	N=10006	N=2683		N=9464	N=7372	N=2092	
Gender					0.556				0.851				0.015
	Male	5421	4072	1349		6265	4936	1329		4493	3451	1042	
	Female	6018	4549	1469		6424	5070	1354		4971	3921	1050	
Age (mean, years)		59.53	59.56	59.45	0.600	61.00	61.02	60.95	0.738	60.58	59,88	63,03	p<0.001
Education					0.070				0.867				0.311
	Never attended school	3225	2475	750		3297	2615	682		1744	1347	397	
	Primary school and lower	4653	3465	1188		4952	3889	1063		4281	3330	951	
	Junior middle school and lower	2314	1726	588		1579	1245	334		2296	1818	478	
	Senior middle school and lower	1247	955	292		2861	2257	604		1143	877	266	
Residential status					0.449				0.808				p<0.001
	Rural	9250	6985	2265		9411	7426	1985		7105	5605	1500	
	Urban	2189	1636	553		3278	2580	698		2359	1767	592	
Marital status					0.046				0.277				p<0.001
	Married / Cohabiting	9916	7509	2407		11026	8671	2355		8361	6572	1789	
	Divorced / Widowed / Separated	1431	1041	390		1583	1269	314		1057	766	291	
	Never married	92	71	21		80	66	14		46	34	12	
Log(Income) (mean,□)		3.73	3.74	3.73	0.403	3.76	3.76	3.76	0.848	3.86	3,88	3,81	p<0.001
Health status					0.448				0.926				p<0.001
	Very good	350	271	79		789	629	160		1276	1130	146	
	Good	1321	984	337		1382	1094	288		1239	1067	172	
	Fair	3654	2731	923		3988	3153	835		4698	3701	997	

Mental health	No	8257	6202	2055	0.611	8460	6676	1784	0.081	6848	5561	1287	p<0.001
	No depressive symptoms	6720	5053	1667		7784	6099	1685		5950	4766	1184	
	Depressive symptoms	4719	3568	1151		4905	3907	998		3514	2606	908	
Diabetes					0.535				0.006				p<0.001
	Yes	659	490	169		745	558	187		1184	592	592	
	No	10780	8131	2649		11944	9448	2496		8280	6780	1500	
CVD					0.156				0.004				p<0.001
	Yes	1553	1148	405		1718	1310	408		2053	1158	895	
	No	9886	7473	2413		10971	8696	2275		7411	6214	1197	
BMI					0.090				p<0.001				p<0.001
	Underweight	811	596	215		752	583	169		485	430	55	
	Normal weight	7125	5385	1740		7476	6016	1460		5712	4787	925	
Drink	Overweight	2941	2196	745	p<0.001	3785	2905	880	p<0.001	2795	1915	880	0.115
	Obese	562	444	118		676	502	174		472	240	232	
Smoke	Drink more than once a month	2762	2153	609	0.605	3406	2774	632	0.700	2581	2040	541	0.002
	Drink but less than once a month	890	710	180		1121	949	172		753	597	156	
	Non-drinker	7787	5758	2029		8162	6283	1879		6130	4735	1395	
Night sleep duration	Yes	4572	3434	1138	0.586	5646	4461	1185	0.743	4048	3092	956	0.001
	No	6867	5187	1680		7043	5545	1498		5416	4280	1136	
https://preprints.jmir.org/preprint/38782	<6	3335	2528	807	0.001	3887	3065	822	0.001	3148	2376	772	0.001
	6-7	2356	1795	561		2620	2061	559		2185	1723	462	
	7-8	2207	1662	545		2250	1770	480		1661	1316	345	
	8-9	2393	1781	612		2593	2035	558		1736	1390	346	
	>9	1148	855	293		1339	1075	264		734	567	167	

30-60	994	762	232	1396	1075	321	1027	794	233
60-90	2717	2101	676	3480	2697	783	2236	1714	522
>90	1416	1073	343	1708	1307	401	1663	1243	420

N.B. ^a: outcomes of Chi-square test and independent one sample t-test.

Association between midday napping and hypertension

Midday napping was found to be positively correlated with suffering from hypertension. In 2011 and 2015, after being adjusted by other control variables, groups of napping, considering the napping duration, i.e., 0-30, 30-60, 60-90, and >90 minutes, were all positively correlated with suffering from hypertension. Napping 60-90 minutes had the greatest ORs (2011: OR, 1.705, CI, 1.346~2.159; 2015: OR, 1.494, CI, 1.227~1.818) compared to non-nappers. In 2018, except for the group napping 0-30 minutes, other groups of participants were all positively correlated with hypertension and the greatest OR came from the group napping 30-60 minutes (OR, 1.223, CI, 1.016~1.473). See figure 1.

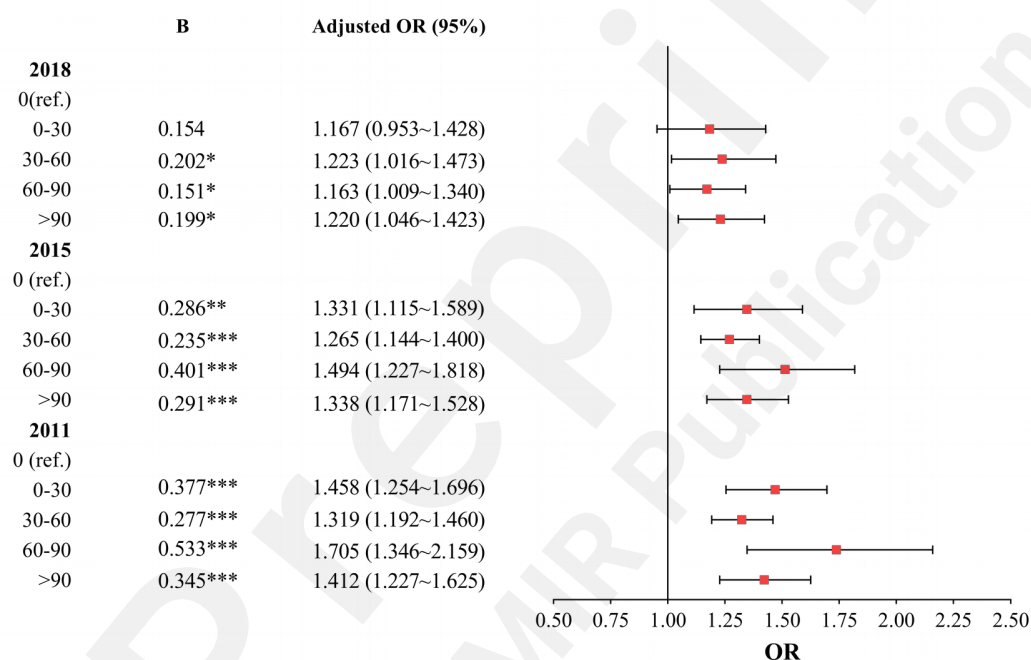


Figure 1. Influence of midday napping on hypertension in different years

N.B. The horizontal line at the end of each line represents the 95% confidence interval, the square in the middle line represents the OR value, and the line segment intersects with the middle vertical line (=1), which means that the result is not significant ($P > 0.05$). Non-intersection means that the result is significant ($P < 0.05$). Unit: minute

***: $P < 0.001$; **: $P < 0.01$; *: $P < 0.05$

From the longitudinal perspective, the ORs of each group of nappers decreased from 2011 to 2018. ORs of napping 60-90 minutes decreased the most, from 1.705 in 2011, to 1.338 in 2015, and to 1.163 in 2018. The ORs of napping over 90 minutes went down from 1.412 in 2011 to 1.220 in 2018. The ORs of napping 30-60 minutes

decreased from 1.319 in 2011 to 1,223 in 2018. Lastly, the ORs of napping 0-30 minutes decreased from 1.458 in 2011 to 1.331 in 2015. See Figure 2.

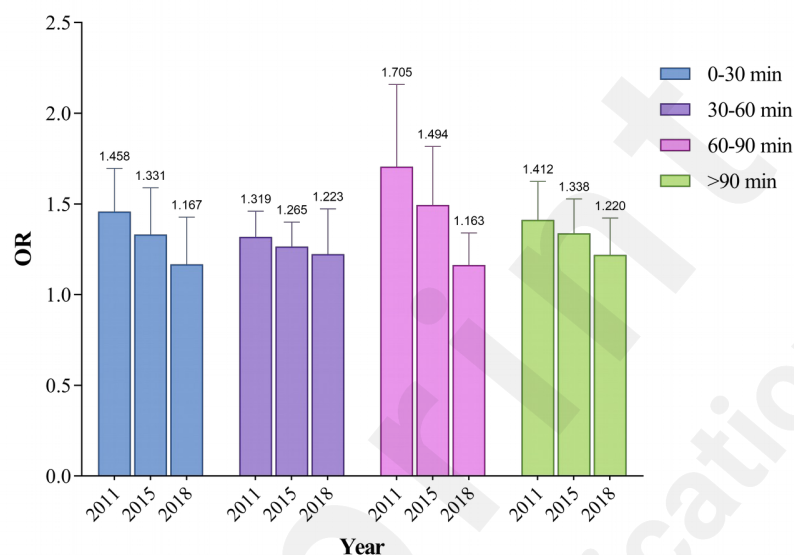


Figure 2. Trajectories of ORs from 2011 to 2018.

N.B. The OR of napping 0-30 minutes was not significant in 2018.

Mediation effect of BMI

The data revealed that the mediation effect of BMI did exist in 2015 and 2018, but not in 2011. In 2015, the total effect of midday napping was found to be significant on hypertension (path c: $B=0.012$, $p<0.001$). Midday napping had a positive effect on BMI in path a ($B=0.011$, $p<0.01$) and BMI had a positive effect on hypertension in path b ($B=0.022$, $p<0.001$). In path c', the effect of midday napping was also significant ($B=0.012$, $p<0.001$), so the BMI was identified to be a partial mediator. The mediation effect was 0.000242, with a ratio of 2.01% to the total effect. In 2018, the coefficients in path a, path b and path c were also found to be significant and the mediation effect of BMI was also identified to be partial, reaching 0.003058. The ratio of the mediation effect over the total effect raised to 23.52%. The details are given in Table 3.

Table 3. Coefficient (B) in testing model of mediation effect

Year	X-Y(path c)	X-M(path a)	(X+M)-Y	
			X(path c')	M(path b)
2011	0.018***	-0.023***	0.017***	-0.005

2015	0.012***	0.011**	0.012***	0.022***
2018	0.013***	0.022***	0.010***	0.139***

N.B. B, unstandardized coefficient; X, Midday napping time; M, BMI; Y, Hypertension.

path c, regression between X and Y.

Path a, regression between X and M.

Path c', regression between X and Y with M controlled.

Path b, regression between M and Y with X controlled.

***: $P < 0.001$; **: $P < 0.01$

Discussion

This cross-sectional study found midday napping positively associated with hypertension among three sectional samples in China, indicating that midday napping may represent a potential causal risk factor. Meanwhile, the ORs of various napping duration decreased over time. The BMI was found to be a partial mediator between midday napping and hypertension.

Although napping has long been regarded as a healthy habit, this study suggests that it may be a potential risk factor for hypertension. Evidence from the UK Biobank [46] and cohort studies in China [32] also supported the results of this study. A meta-analysis concluded that the pooled RR of hypertension in nappers was 1.13 based on nine observational studies [47]. However, disparities between this study and existing literature also exist, reminding us of a cautious interpretation of the results. Some other studies found midday napping to have a protective effect for habitual nappers compared with those who never napped [48] or could decrease the risk of hypertension in specific napping durations [49, 50], which are contradictory to the results in this study. Meanwhile, in this study, different durations of midday napping were all found to be positively associated with hypertension (except napping for 0-30 min in 2018). However, the associations of midday napping duration with hypertension differed in various studies. For example, the significantly increased odds for hypertension were only found in participants napping over 90 minutes in some studies [34] [31]. Another cohort study conducted in China including 13706 participants found no significant associations of napping for less than 30 min with hypertension [35]. The inconsistency might be explained by study designs and samples, different characteristics of participants, the disparity in included confounders, and the measurements of napping behaviors and other confounders across studies. Therefore, it is important to be cautious about the results and long-term follow-up and experimental studies are needed to determine the exact impacts of midday napping.

From 2011 to 2018, decreases in ORs were seen in different napping durations (napping 0-30 min: 1.458 to 1.331; napping 30-60 min: 1.319 to 1.223; napping 60-90 min: 1.705 to 1.338; napping over 90 min: 1.412 to 1.220). Some speculations were given to help understand the results. First, there were only four variables significant (including education, marital status, drinking, and napping duration) in the regression model of 2011. But the corresponding number was eight when it came to 2018 (including gender, age, health status, ADL, diabetes, CVD, BMI, and napping duration). The increasing correlation between significant variables might decrease the value of ORs. Second, the great socioeconomic and environmental transformations related to hypertension during 2011~2018 were not controlled in our study, such as dietary patterns, exposure to fine particulate matter (PM 2.5), built environment factors, and some other confounders [51-53]. Third, the association of midday napping with hypertension might be moderated by other variables such as physical conditions and night sleep duration [48, 54]. Therefore, current evidence was not enough to conclude that the impact of midday napping got weakened. And it can only be explained after figuring out the hidden specific mechanism. However, the results also deserve our attention because they indicated the possibility that the potential risk of midday napping might be mitigated or even eliminated if we can control other confounders well.

The mediation effect of BMI was identified in this study. Some previous studies found that nappers were more likely to be overweight or obese [33, 55, 56]. So, it could be inferred that midday napping contributes to hypertension by elevating the risk of obesity or overweight, which is an acknowledged risk factor for hypertension [57-59]. However, the ratio of mediation effect over the total effect was 2.01% in 2015 and 23.52% in 2018, indicating the existence of other mediators. It was suggested that midday napping could result in sympathetic surge and an increase in nighttime cortisol, which elevated blood pressure [25]. Midday napping was also regarded as a symbol of sleep apnea, and it was concluded that it was the sleep apnea, not the napping itself, that resulted in cardiovascular diseases [60]. Furthermore, prolonged midday napping may have an impact on the duration and quality of evening sleep [61]. All these factors can indirectly increase the risk of hypertension.

To prevent hypertension, prolonged midday napping should be avoided and actions related to losing weight such as more physical activities and a balanced diet are also needed, especially for nappers. Additionally, further research is needed to define the vulnerable population and develop corresponding interventions.

Conclusion

In this study, it was found that midday napping was positively associated with hypertension in Chinese middle-aged and older people. Though the causal effects were hard to prove, BMI was found to play the role of mediator. Therefore, avoiding prolonged midday napping and actions to maintain a normal BMI level are recommended. With the deepening of research, the specific mechanism of interaction between midday napping and hypertension deserves more attention. Besides, it's also of great significance to investigate other implications of midday napping considering its high prevalence.

Limitations

In this study, there were some limitations that should be mentioned for cautious interpretation. First, despite the positive correlation observed, the regression model that was employed as well as the cross-sectional study design were not robust enough to conduct the causal inference, which weakened the evidence. Second, the use of self-reported midday napping duration and some other variables might introduce recall bias. Third, although some confounders were adjusted in the model, potential residual covariates might remain due to the absence of information such as genetic factors, a family history of hypertension, bio-markers and environmental factors. Besides, time-dependent covariates were not included in our study, which made comparisons across years difficult. Fourth, all participants were aged ≥ 45 years old from China. Therefore, it remains uncertain whether the conclusion can be applied to other age groups. Additionally, though we added the variable of night sleep duration as a control variable, the potential interaction effect of midday napping and night sleep was not analyzed in this study.

Acknowledgement

Funding: This work was funded by National Key R&D Program of China under Grant No.2020YFC2006500; and National Natural Science Foundation of China under Grant.72004073; and Chinese Ministry of Education of Humanities and Social Science project under Grant NO: 20YJC630134. *Author contributions:* Dongfeng, Tang: conception of the study design, data analysis, interpretation of the data, writing of the manuscript. Yiheng, Zhou: conception of the study design, data analysis, writing of the manuscript. Chengxu, Long: conception of the study design, writing of the manuscript. Shangfeng Tang: conception of the study design, writing of the manuscript.

Conflicts of Interest

None declared.



Abbreviations

CHARLS: China Health and Retirement Longitudinal Study



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